

Samuel Alipour-fard

THEORETICAL PHYSICIST · PH.D., MIT

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Education

Massachusetts Institute of Technology

PH.D. IN THEORETICAL PHYSICS (NUCLEAR & PARTICLE)

Cambridge, MA

Aug. 2019 – May 2025

- Thesis: *Particles Inside Particles: The Flow of Energy in Quarks, Gluons, and Jets*, advised by Jesse Thaler.
- Awarded a 2019 MIT Graduate Fellowship. Cumulative GPA: 4.0/4.0.

University of California, Santa Barbara

B.S. IN PHYSICS, COLLEGE OF CREATIVE STUDIES (CCS)

Santa Barbara, CA

Sep. 2015 – Jun. 2019

- Graduated with highest academic honors and with distinction in the major. Cumulative GPA: 4.0/4.0.
- Arnold Nordsieck Award (to a graduating senior of exceptional research promise); UCSB Regents Scholarship, all four years.
- Research advisor: Nathaniel Craig. Academic advisor: Tengiz Bibilashvili.

Research Experience

Massachusetts Institute of Technology

GRADUATE RESEARCHER, THEORETICAL HIGH-ENERGY PHYSICS (ADVISOR: JESSE THALER)

Cambridge, MA

Aug. 2019 – May 2025

- Built the analytic and probabilistic foundations of energy-correlator observables, and derived from scratch the mathematics underlying parton-shower simulation algorithms.
- Designed new energy correlators that make the low-energy structure of high-energy jets both analytically tractable and cheap to compute; introduced a continuous jet-grooming method (PIRANHA).
- Wrote and released the Monte-Carlo and simulation software behind these results, and used it to test how the observables behave under detector noise and pileup contamination.
- Used the Energy Mover's Distance, a metric on the space of radiation patterns, to reproduce standard collider observables from a geometric starting point.

Engineering Quantum Systems (EQuS) group, MIT (W. Oliver)

CO-AUTHOR, SUPERCONDUCTING-QUBIT NOISE STUDY

Cambridge, MA

2024

- Helped with Lindbladian-dynamics simulations in Mathematica for a study of qubit-state purity oscillations under anisotropic transverse noise; co-author on the resulting *Phys. Rev. A* publication (see Publications).

University of California, Santa Barbara

UNDERGRADUATE RESEARCHER, HIGH-ENERGY PHYSICS (ADVISOR: NATHANIEL CRAIG)

Santa Barbara, CA

Jan. 2017 – Jul. 2019

- Adapted MadGraph, Pythia 8, and Delphes to simulate displaced decays and long-lived particles (LLPs) in high-energy collisions.
- Studied the reach of current and proposed colliders (LHC, CEPC, ILC, FCC-ee, CLIC) for detecting LLPs; work published across three papers (see Publications) and presented in talks by N. Craig and S. Koren.
- Roig Summer Undergraduate Research Fellow (2018); Arnold Nordsieck Award (2019).

University of California, Santa Barbara

UNDERGRADUATE RESEARCHER, BIOPHYSICS (ADVISOR: GREG HUBER)

Santa Barbara, CA

Apr. 2016 – Jun. 2018

- Modeled diffusion on “parking-garage” (Terasaki-ramp) topologies of the kind found in the rough endoplasmic reticulum, writing code to simulate random walks on these surfaces.

Technical Skills

Programming

Python (scientific computing, PyTorch/TensorFlow), C++, Mathematica, Bash, $\text{\LaTeX}$

Methods

Monte-Carlo methods and numerical simulation, statistical inference, analytic computation, Git

Physics & Math

Quantum field theory and effective field theory, statistical mechanics, probability and information theory, perturbative QCD

Software written

Public research code (github.com/samcaf): JetMonteCarlo, ResolvedEnergyCorrelators, ShowerPy

Languages

English (native), Spanish

Publications

Probing Jet Quenching Dynamics with Higher-Point Energy Correlators

In prep.

S. Alipour-fard, A. Budhraj, A. Kudinoor, J. Thaler, W. Waalewijn.

Particles Inside Particles: The Flow of Energy in Quarks, Gluons, and Jets

2025

S. Alipour-fard. Ph.D. thesis, Massachusetts Institute of Technology.

Energy Correlators Beyond Angles

2025

S. Alipour-fard, W. Waalewijn. JHEP **07** (2025) 043. arXiv:2501.17218.

New Angles on Energy Correlators

2025

S. Alipour-fard, A. Budhraj, J. Thaler, W. Waalewijn. Phys. Rev. Lett. **134** (2025) 231902. arXiv:2410.16368.

Qubit-State Purity Oscillations from Anisotropic Transverse Noise

2025

D. A. Rower *et al.* (incl. S. C. Alipour-Fard), Engineering Quantum Systems group, MIT. Phys. Rev. A **111** (2025) 032420. arXiv:2409.12303.

Pileup and Infrared Radiation Annihilation (PIRANHA): a Paradigm for Continuous Jet Grooming

2023

S. Alipour-fard, P. T. Komiske, E. M. Metodiev, J. Thaler. JHEP **09** (2023) 157. arXiv:2305.00989.

The Second Higgs at the Lifetime Frontier

2020

S. Alipour-fard, N. Craig, S. Gori, S. Koren, D. Redigolo. JHEP **07** (2020) 029. arXiv:1812.09315.

Long Live the Higgs Factory: Higgs Decays to Long-Lived Particles at Future Lepton Colliders

2019

S. Alipour-fard, N. Craig, M. Jiang, S. Koren. Chin. Phys. C **43** (2019) 053101. arXiv:1812.05588.

The CLIC Potential for New Physics

2018

J. de Blas *et al.* (incl. S. Alipour-fard). CERN Yellow Reports: Monographs, CERN-2018-009-M. arXiv:1812.02093.

Selected Talks

Continuous jet grooming and energy correlators (PIRANHA)

INVITED AND CONTRIBUTED TALKS

2021 – 2024

- Invited: ATLAS Jet Definitions subgroup, CERN (2021); LHCP (2022); “Jet Physics: From RHIC/LHC to EIC,” Stony Brook Center for Frontiers in Nuclear Science (2022).
- Contributed: Boost 2021 (remote), Boost 2022 (Hamburg), Boost 2024 (Genova).

Particles Inside Particles: The Flow of Energy in Quarks, Gluons, and Jets

MIT

PH.D. THESIS DEFENSE

2025

- Public talk introducing the physics of quarks and gluons and my work on removing detector noise and contamination from particle data. Recording: youtube.com/watch?v=ZI27iiX8qFM.

Long-lived particles; supersymmetric QM; generalized functions

UC Santa Barbara

UNDERGRADUATE RESEARCH AND INDEPENDENT SEMINARS

2017 – 2018

- Talks and posters on long-lived particles at current and future colliders (UCSB Undergraduate Research Symposium, CCS RACA-CON; recordings at KITP).
- Organized and led interactive student seminars on supersymmetric quantum mechanics and on generalized functions.

Teaching

Massachusetts Institute of Technology

Cambridge, MA

TEACHING ASSISTANT, DEPARTMENT OF PHYSICS

Aug. 2020 – May 2025

- TA for graduate courses in Quantum Field Theory (8.323), Effective Field Theory (8.851), Supersymmetric QFT (8.831), and General Relativity (8.962); and for undergraduate Electromagnetism (8.02), String Theory (8.251), and the Physics of Energy (8.21).
- Delivered one-third of the lectures for the graduate SUSY QFT course during the instructor's absence; designed problems, ran recitations, and helped put the EFT course on the MITx platform. Perfect TA evaluations in 8.251.

University of California, Santa Barbara

Santa Barbara, CA

LEARNING ASSISTANT, LEARNING ASSISTANT PROGRAM

Sep. 2017 – Jul. 2019

- Supported upper-division Quantum Mechanics, Classical Mechanics, and Thermodynamics & Statistical Mechanics, and lower-division Electrodynamics. This work contributed to N. Craig's 2018 Cottrell Scholar Award.

Boulder High School

Boulder, CO

TEACHING ASSISTANT, CALCULUS III

Aug. 2014 – Jan. 2015

- Helped high-school students learn multivariable calculus, under teacher Akili Obeka.

Honors, Awards, and Scholarships

2020 **Finalist**, Paul and Daisy Soros Fellowships for New Americans

New York, NY

2019-2020 **Recipient**, MIT Graduate Fellowship

Cambridge, MA

2019 **Bachelor of Science**, graduated from UCSB with highest academic honors, and distinction in the major

Santa Barbara, CA

2019 **Recipient**, Arnold Nordsieck Award, granted to a graduating senior with exceptional research promise

Santa Barbara, CA

2017 **Recipient**, Roig Summer Undergraduate Research Fellowship

Santa Barbara, CA

2017-2019 **Recipient**, Regents Scholarship

Santa Barbara, CA

Service & Outreach

USA Young Physicists' Tournament (USAYPT)

JUDGE

2020 – 2025

- Judged four editions of this international tournament for middle- and high-school teams (Exeter NH, remote, Raleigh NC, Andover NH), with teams from the US, China, Georgia, and Tunisia.

USA Physics Olympiad

GRADER AND COACHING TEAM

2022

- Graded and discussed problems with the US coaching team for the 2022 USAPhO.

MIT graduate admissions

Cambridge, MA

FOUNDING MEMBER & APPLICATION REVIEWER

2020 – 2022

- Founding member of Graduates Advising Graduate Admissions; reworked application and essay prompts to broaden access. Reviewed applications for the MIT Summer Research Program (MSRP).

Feynman Club, University of California, Santa Barbara

Santa Barbara, CA

FOUNDER

2018

- Ran a student club (with T. Bibilashvili) teaching quantum mechanics through the path-integral approach, working problems together in a conversational style.